

Automated AI Assisted Trading

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ABSTRACT

Stock trading strategies play a critical role in investment. However, it is challenging to design a profitable strategy in a complex and dynamic stock market. Ever since the stock market was established, investors have been trying to determine a strategy capable predicting such a market. Many solutions have been proposed including support vector machines, fuzzy classifiers and Bayesian models. This paper will focus on the application of deep reinforcement learning algorithms in producing a profitable trading strategy. Specifically, five actor-critic algorithms will be tested. These are Deep Deterministic Policy Gradient (DDPG), Advantage Actor Critic (A2C), Soft Actor Critic (SAC), Twin Delayed Deep Deterministic Policy Gradient (TD3), and Proximal Policy Optimization (PPO). These algorithms will be evaluated and compared to a number of metrics such as annual cumulative return, Sharpe Ratio, and DOW Jones Industrial Average Index. The Results of these tests will be combined into an ensemble trading strategy and applied to a live trading environment for real time testing.

1 INTRODUCTION

1.1 Problem Definition

The stock market is a collection of exchanges in which shares of publicly held companies are bought and sold, with the intention of making a profit. For as long as the market has existed investors have tried to determine strategies which could guarantee a profitable outcome. Many prominent machine learning techniques have been applied to this problem include, Bayesian models [1], fuzzy classifiers [2], and support vector machines [3], as well as qualitative indicators such as financial reports, social media postings, and global affairs.

Over the last decade researchers have invested their time into new algorithmic approaches such as, deep Reinforcement Learning. Reinforcement Learning is a type of machine learning that has been applied successfully to many other fields, including recommendation systems [4], energy management [5, 6], phishing scam detection [7], and drone navigation [8]. With regards to the financial market, reinforcement learning has shown promising results, outperforming traditional trading methods such as ARIMA [9].

1.2 Aims and Objectives

This project aims to answer the question, does the class of reinforcement learning algorithms show potential to produce a profitable trading strategy, and how can such a strategy be implemented into an automated agent. To achieve this aim, the following objectives were set.

- (1) Implement, optimize, and test a number of reinforcement learning algorithms with regards to profitable and stable income when trading on the stock market.
- (2) Following the testing in Objective 1, test and create an ensemble trading strategy.
- (3) Implement said trading strategy into an automated agent.

2 BACKGROUND

The field of market predication is one that has been studied extensively over the years, producing a depth of knowledge needed to understand the space.

2.1 Data Format

Market data tends to always follow the same format, where for each time step five properties are provided. These are:

- **Open Price:** Refers to the price of an asset at the beginning of the time step.
- **High Price:** Refers to the highest prices an asset reached within a time step.
- **Low Price:** Refers to the lowest price an asset reached within a time step.
- **Close Price:** The final price of the asset within the time step.
- **Volume:** The amount of shares that were traded during the time step.

This format is referred to as OHLCV. In addition to this, some datasets also proved an indicator for which day of the week the time step occurred on. This is referred to as OHLCVD.

2.2 Technical Indicators

Technical indicators are heuristic or pattern-based statistical models produced by the data points described in Section 2.1. Traditionally, traders would use these indicators to model complex patterns in the hopes of determining the direction of an asset. Some examples of technical indicators are, relative strength index (RSI), simple moving average (SMA), and Bollinger Bands.

RSI [10] is a momentum indicator which measures the speed and magnitude of an assets recent price changes. The term momentum indicator refers to a technical indicator acting over a period of n days, measuring the strength or weakness of an asset. RSI is used to determine overvalued or undervalued conditions in the price of the asset, and is defined as:

$$RSI = 100 - \left(\frac{100}{1 + \frac{n_{up}}{n_{down}}} \right) \quad (1)$$

Where n_{up} is the average gain over a period of n time steps, and n_{down} is the average loss over n time steps.

SMA is used to calculate the average of a selected range of prices, and is used to determine the stability of an asset. It is defined as:

$$SMA = \frac{A_1 + A_2 + A_3 + \dots + A_n}{n} \quad (2)$$

Where A_n is the price of an asset at period n , and n is the number of total periods.

Similarly, Exponential Weighted Moving Average (EWMA) [11] is a weighted version of SMA, giving more importance to more recent time steps. The relevance of each time step decreases exponentially the further away from the present it gets. EWMA is defined as:

$$\sigma_n^2(EWMA) = \lambda \sigma_{n-1}^2 + (1 - \lambda) U_{n-1}^2 \quad (3)$$

where $\sigma_n^2(EWMA)$ is the EWMA variance at time step n , λ is the degree of weighting, $\sigma_{n-1}^2(EWMA)$ is the variance at time step $n - 1$ and U_{n-1}^2 is the squared EWMA return at time step $n - 1$.

With regards to machine learning, technical indicators are used to identify trends and likely of short or long term change.

2.3 Risk Assessment

Financial risk refers to any uncertainty with respect to your investments that has the potential to negatively impact your financial welfare [12]. The following are some of the main tools used by investors to quantify risk.

2.3.1 Sharpe Ratio. The Sharpe ratio is one of the most popular metrics for measuring risk. To do this the ratio considers a risk free asset as a comparison metric. This asset is often taken as the U.S. treasury bonds over the same duration of the investment [13]. The Sharpe ratio is defined as:

$$S_a = \frac{E[R_a - R_b]}{\sqrt{\text{var}[R_a - R_b]}} \quad (4)$$

Where R_a is the asset return, R_b is the risk-free return, E is the expected value, and var is the variance over the respective excess.

2.3.2 Sortino Ratio. The Sortino ratio [14] was developed off the Sharpe Ratio. Sortino adapted the algorithm to penalize negative outcomes quadratically. Formally the Sortino Ratio is given as

$$S = \frac{R - T}{DR} \quad (5)$$

Where R is the average return, T is the minimum accepted return, and DR is the downside risk.

2.3.3 Turbulence. While not explicitly a measure of risk, turbulence is a metric to determine the volatility of the market, and whether the conditions are too unpredictable to continue trading [15]. Typically, turbulence is used as a threshold whereby once its value exceeds the predetermined amount all algorithmic trading would be stopped and assets are sold. This metric is time dependent and is defined as

$$\text{turbulence}_t = (x_t - \mu) \Sigma^{-1} (x_t - \mu)^T \quad (6)$$

Where x_t is the return of the asset at time-step t , μ is the mean return, and Σ is the covariance of the returns.

The Chicago Board Options Exchange Volatility Index (VIX) is one of the most popular measurements of turbulence. It reflects instability over the last 30 days.

2.4 Reinforcement Learning

Reinforcement Learning (RL) is a type of machine learning which focuses on interactions between an agent and the environment. The main components of an RL network are the environment, agent, reward, and policy. Figure 1 gives an overview of the RL interaction loop.

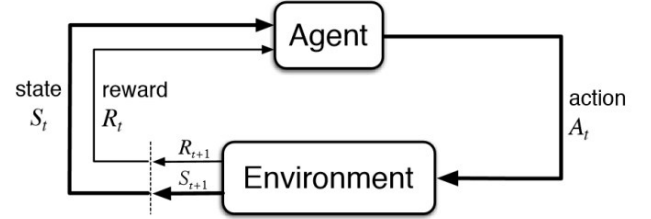


Figure 1: The Reinforcement Learning Interaction Loop [16]

In a time step of the RL iteration loop, the agent obtains an observation (S_t) containing a reward (R_t). The agent uses this information to subsequently select an action from the action space. This action is applied to the environment, creating a new state and reward denoted as S_{t+1} and R_{t+1} respectively. This is repeated for all the time steps with the purpose of maximising the cumulative reward.

The mathematical formulation of RL problems is called the Markov Decision Process (MDP). This is defined as a class of stochastic sequential decision processes in which the cost and transition functions depend only on the current state of the system and the current action [17]. Therefore, the current state in a MDP is only affected by the previous state. Solutions to a MDP are defined as sets of actions that the agent requires to reach a goal [18], or policies. The set of all states, and the set of all actions in a MDP are denoted as S and A respectively. A policy is represented as $\pi(s) \rightarrow a$, where $a \in A$ and $s \in S$.

RL algorithms are typically value based or policy based. Value based algorithms recall and update a collection of rewards for each action in a state. In non-deep learning approaches this may be just a table (Q-table), while in deep learning approaches this would be a neural network. Therefore, value-based approaches learn state-action values to determine the best action in a given state. To encourage exploration, an Epsilon Greedy [19] policy selects either the best action or a random action with probability $1 - \epsilon$. An algorithm that makes use of this approach is Q-Learning [20]. Policy based approaches learn directly the stochastic policy function that maps each state to an action. The policy will produce probabilities for each action in a given state, and an action will be chosen based on those probabilities. This means that, unlike value-based approaches, policy-based approaches do not require an exploration function. One such example of this is policy gradient [21].

2.4.1 Actor Critic. Actor Critic algorithms are an adaptation of the policy gradient with baseline algorithm, where a temporal difference approach replaces policy gradients Monte Carlo approach. The system makes use of two function approximates. These are the actor - a policy-based function parameterized by θ : $\pi_\theta(s, a)$ - and the critic, a value-based function parameterized by w : $\hat{q}_w(s, a)$.

In an actor-critic iteration step, the actor takes in a state S_t and outputs an action A_t based on its policy. This action is fed into the value-based critic function along with the state to compute the Q-value $\hat{q}_w(s_t, a_t)$. Similarly, the action is passed through the environment, where a new state S_{t+1} and reward R_{t+1} are returned. These values are used to update the two function approximations. Following is the equation used to update the actor:

$$\Delta\theta = \alpha \nabla_{\theta}(\log \pi_{\theta}(s, a)) \hat{q}_w(s, a) \quad (7)$$

Where, $\Delta\theta$ is the change in policy parameters, and $\hat{q}_w(s, a)$ is the action value estimate produced by the critic function.

This update allows the actor to produce the next action A_{t+1} given the state S_{t+1} . This is used to update the critic network using the following formula:

$$\Delta w = \beta(R(s, a) + \gamma \hat{q}_w(s_{t+1}, a_{t+1}) - \hat{q}_w(s_t, a_t)) \nabla_w \hat{q}_w(s_t, a_t) \quad (8)$$

The dynamic between the actor and critic networks affects the variance of the system without changing its direction. This results in a more reliable algorithm [21].

2.5 Actor Critic Algorithms

The following is a list of state-of-the-art actor critic algorithms that serves as an example of the direction research is going in.

2.5.1 Advantage Actor Critic (A2C). A2C [22] is a standard actor-critic algorithm which improves on the policy gradient update. This is done by utilizing an advantage function to reduce the variance in the policy gradient. The advantage function is computed alongside the value function in the critic network. The addition of the advantage function gives the algorithm the ability to determine not only how good an action was, but how good an action can be. The objective function for the A2C algorithm is given as.

$$\nabla J_{\theta}(\theta) = \mathbb{E} \left[\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) A(s_t, a_t) \right] \quad (9)$$

Where $\pi_{\theta}(a_t | s_t)$ is the policy network and $A(s_t, a_t)$ is the advantage function.

2.5.2 Proximal Policy Optimization (PPO). PPO is an algorithm part of the policy gradient methods family. It is based on the trust region policy optimization (TRPO) algorithm [23], and works by alternating between sampling data through interaction with the environment, and optimizing a "surrogate" objective function using stochastic gradient ascent [24]. The Proximal Policy Optimization update function is defined as:

$$L^{CLIP}(\alpha) = \hat{E}_t = \min[r_t(\alpha) \hat{A}_t, \text{clip}(r_t(\alpha), 1 - \epsilon, 1 + \epsilon) \hat{A}_t] \quad (10)$$

Where $r_t(\alpha) \hat{A}_t$ corresponds to the update function used in TRPO [23], the clip function ensures that updates are bound between $1 \pm \epsilon$ (where ϵ is a small value typically between 0.1 – 0.3, used to set the size of the clip), and the minimum function ($\min$) ensures that learning updates are not excessively large and therefore unstable.

Additionally, $\hat{A}_t$ acts as a switch between good and bad updates. This is depicted in figure 2 below.

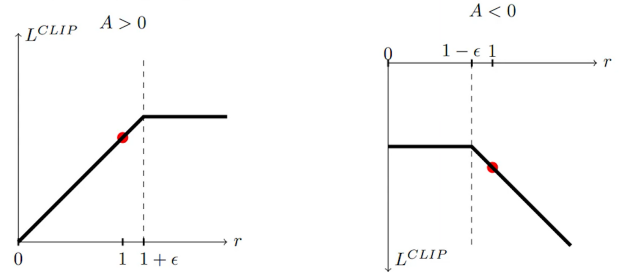


Figure 2: Effects of A in PPO update function

3 LITERATURE REVIEW

The field of market prediction has been around for as long as the market itself, with the main question being, whether it is possible to predict such a volatile environment. This has lead some researchers to suggest that the financial markets follow the structure of a random walk [25, 26]. This entails that the market cannot be predicted, and that the current state of the market is the best prediction that will exist. On the contrary, the market has also been linked to the theory of concept drifts [27]. Concept drifts refer to a system's ability to change its underlying key components and subsequently its behaviour over time. While none of this research was conclusive, it shows the uncertainty revolving around the dynamics of a market predicting model.

Over the years a number of different techniques were tested. One classic stock price forecasting method is Autoregressive Integrated Moving Average (ARIMA) [9]. This model uses OHLCV data to predict future values for time-series data, but can not be used if the data contains seasonal components. Seasonal components are defined as repetitive cycles found in the data. To model such data, advanced ARIMA (SARIMA) was proposed [28]. SARIMA is an effective financial forecasting model for short to medium range data.

These traditional models have been used effectively on stationary time series data. However, applying such models to a live trading environment has not been successful [29]. This is because it is impossible to predict whether incoming new data is stationary time series data. To solve this, some researches have applied deep learning to the problem [24, 30, 31].

Deep learning is a family technique of machine learning which makes use of neural networks. This has been combined with reinforcement learning to create Deep Reinforcement Learning (DRL), the main contributor to which is actor-critic, as outline in section 2.4.1. One approach, as can be seen in [32], applies an ensemble strategy combining three actor-critic algorithms to try create a stable model capable of predicting non-stationary time-series data. The chosen algorithms were PPO, A2C, and SAC. The algorithms were trained using daily opening and closing prices, a stock sentiment indicator ARBR, and twenty four technical indicators. The indicators were passed through a PCA algorithm, reducing them to a 20-dimensional vector. Table 1 depicts the results of this experiment.

The results of this experiment concluded that, although the cumulative return of the PPO algorithm was greater than that of the ensemble strategy, the ensemble strategy outperformed the PPO

Table 1: PPO, A2C, Sac Ensemble Strategy Results

2017/01/01 - 2021/04/25	Ensemble	PPO	A2C
Cumulative Return	71.92%	81.99%	50.73%
Annual Return	17.98%	20.5%	12.68%
Annual Volatility	9.7%	14.8%	11.3%
Sharpe Ratio	1.41%	1.12%	1.15%
Max Drawdown	-9.5%	-25.3%	-18.4%
2017/01/01 - 2021/04/25	SAC	MACD	Min-variance
Cumulative Return	49.23%	29.32%	24.27%
Annual Return	12.31%	7.33%	6.07%
Annual Volatility	14.9%	18.3%	20.1%
Sharpe Ratio	0.78%	0.42%	0.45%
Max Drawdown	-14.8%	-38.8%	-34.6%

algorithm. This is due to the higher Sharpe ratio, lower max drawdown, and lower volatility of the ensemble strategy which made it more stable and consistent, enabling it to make more consistent profits. Note that the trading range includes the unstable 2020 year, and part of the 2021 year.

Similarly, [15] depicts the use of an ensemble strategy consisting of PPO, A2C, and DDPG. The ensemble strategy used is described as follows:

Step 1. A growing window of n months is used to retrain each of the algorithms to be used in the ensemble strategy. Each algorithm is re-trained every 3 months.

Step 2. Each agent is validated on a 3-month validation rolling window, and the Sharpe ratio for each is calculated.

Step 3. The best performing agent is used for the next 3-month period. This is done as each algorithm is sensitive to different market paradigms, and therefore may perform better as the market shifts. The results of this paper are depicted in Table 2.

Table 2: PPO, A2C, DDPG Ensemble strategy Results

2016/01/04-2020/05/08	Ensemble	PPO	A2C
Cumulative Return	70.4%	83.0%	60.0%
Annual Return	13.0%	15.0%	11.4%
Annual Volatility	9.7%	13.6%	10.4%
Sharpe Ratio	1.30%	1.10%	1.12%
Max Drawdown	-9.7%	-23.7%	-10.2%
2016/01/04-2020/05/08	DDPG	Min-Variance	DJIA
Cumulative Return	54.8%	31.7%	38.6%
Annual Return	10.5%	6.5%	7.8%
Annual Volatility	12.3%	17.8%	20.1%
Sharpe Ratio	0.87%	0.45%	0.47%
Max Drawdown	-14.8%	-34.3%	-37.1%

The results of this experiment match closely with those of [32], where the ensemble strategy is outperformed by the PPO algorithm in terms of cumulative returns, but gains significant stability in return.

4 METHODOLOGY

4.1 Algorithm Sections

To determine which algorithms would be considered for this research, a number of criteria were considered. First, the algorithms should be ones which have shown prevalence in the field, and therefore have proven themselves as possible contenders for an idealistic trading strategy. The algorithms should also be state-of-the-art in their respective categories. Based on previous work [15, 24, 30–32], the following algorithms were chosen.

- Deep Deterministic Policy Gradient (DDPG)
- Twin Delayed Deep Deterministic Policy Gradient (TD3)
- Soft Actor Critic(SAC)
- Advantage Actor Critic (A2C)
- Proximal Policy Optimization (PPO)

Note, all of the chosen algorithms make use of the actor-critic network structure described in section 2.4.1.

4.2 Objective 1

To implement the previously mentioned algorithms, the FinRL¹ library was used. FinRL is an open-source framework focused on financial reinforcement learning. It can be described as an ecosystem composed of hundreds of financial markets, state-of-the-art algorithms, and financial applications such as portfolio allocation, live trading and cryptocurrencies. Following the outlined scope, objective 1 will make use of FinRLs reinforcement learning implementations and multi stock trading environment.

As a base line, all algorithms were run on an environment containing OHLCVD data (section 2.1), turbulence ceilings, and the VIX index. Some algorithms have also been tested on environments including technical indicators and sets of principle component analysis (PCA) variables. Note these environments are subject to change based on further research and testing. Moreover, the algorithms will be subjected to one of the aforementioned risk analysis metrics described in Section 2.3. This is done to strike a balance between risk taken and speculated returns.

Going forward, testing will be performed on all the algorithms with different environments. The use of PCA will also be investigated further to evaluate the ideal amount of information integrity preserved while also reducing the dimension of the environment.

To ensure optimal hyperparameters, the Optuna package will be used. Optuna is an automatic hyperparameter optimization software which can be applied to any machine learning model. Additionally, k-fold cross-validation will be used in tandem with Optuna to prevent over-fitting more recent data.

4.3 Objective 2

As discussed in Section 3, promising results have been shown when combining multiple actor-critic algorithms into an ensemble strategy. This objective aims to create an ensemble strategy out of the aforementioned algorithms to achieve a more stable model. The approach that will be used for this will follow closely that of [32], where the Sharpe ratio was used as the deciding factor. Testing

¹<https://github.com/AI4Finance-Foundation/FinRL>

of the ensemble strategy will be similar to that of the individual algorithms seen in Section 5.

4.4 Objective 3

As previously mentioned, the FinRL library has support for live trading environments. This will be used in conjunction with a trading platform that supports algorithmic trading such as Alpaca². To test the algorithm on live data a paper trading system will be used.

To connect to the Alpaca Trading Brokerage, the alpaca python package will be used. The package allows users to connect to there account through an API key. Through this connection information such as, market open/close times, account balance, and assets held can all be retrieved. The connection also allows the submission of buy and sell requests.

The development of this project is being produced primarily in python3.

5 EVALUATION

To evaluate the algorithms being used, each algorithm is tested on a year of out-of-sample data. From this, the following metrics are computed.

General Backtesting results: This is a set of information points aimed at giving a general overview of the algorithms' performance. Table 3 shows an example of these results for the A2C algorithm.

Table 3: Results for A2C Backtesting

Start Date	2020-01-02
End Date	2020-12-30
Annual Return	18.008%
Cumulative Returns	18.008%
Annual Volatility	30.238%
Sharpe Ratio	0.70
Calmar Ratio	0.63
Stability	0.36
Max Drawdown	-28.65%
Omega Ratio	1.14
Sortino Ratio	0.95
Skew	NaN
Kurtosis	NaN
Tail Ratio	0.90
Daily Value at Risk	-3.725%
Alpha	0.14
Beta	0.68

Some of the key attributes in this table are the annual return, indicating the average annual return - the Sharpe ratio, Sortino Ratio, and daily value at risk - used to evaluate the risk tolerance of the algorithm - and the maximum drawdown.

Drawdown Periods: Drawdowns are defined as periods of time, where trading is resulting in a loss. Figure ?? and Table 3 below depict these regions for the A2C algorithm over the 2020 year.

This information is cross referenced with the trajectory of the market to ensure that the loss periods of the algorithm are inline

²<https://alpaca.markets/>

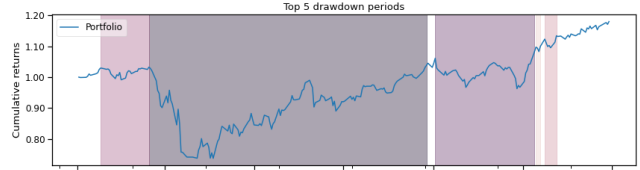


Figure 3: Top 5 Drawdown Periods Graph for A2C

Table 4: Top 5 Drawdown Periods Report for A2c

Index	Net Drawdown in %	Peak Date	Valley Date
1	28.65	2020-02-19	2020-04-03
2	9.30	2020-09-02	2020-10-28
3	3.72	2020-01-17	2020-01-31
4	2.50	2020-11-16	2020-11-20
5	1.33	2020-11-10	2020-11-12

Index	Recovery Date	Duration
1	2020-08-27	137
2	2020-11-09	49
3	2020-01-31	24
4	2020-11-20	7
5	2020-11-12	4

with downwards trajectories in market. If these periods do not align and the maximum drawdown index is of significant value, then further investigation would need to be performed to determine why the algorithm is taking said decisions. The A2C example given here is in line with the turbulent downward trend of the 2020 year.

Risk Assessment: Risk assessment is one of the most important attributes to be measured as it can determine whether a positive result was achieved through accurate decision making or blind luck [33]. For this purpose mainly the Sharpe ratio was used. Figure 4 shows how the value of the Sharpe ratio changed for A2C over the last six month of 2020.

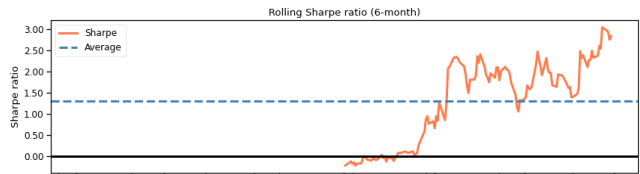


Figure 4: Rolling Sharpe Ratio for A2C

As can be observed, the Sharpe ratio for A2C at the six month mark is below zero. This means that the algorithm was performing worse than the risk free return. Later, the Sharpe value rises well above the value of two and ends with an average value around 1.4. Note that indexes such as the DOW have an average Sharpe ratio value of 0.06.

Algorithm Comparison to DOW index: Another visual indicator that can aid in evaluating the performance of an algorithm is comparing its return to that of an established fund. Figure 5 below

compares the results of A2C to the DOW Jones Industrial Average index.

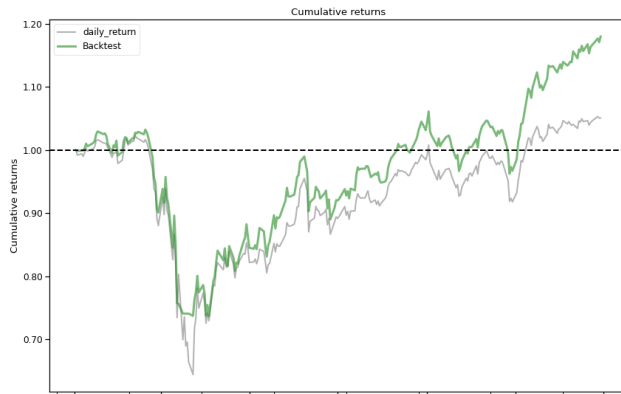


Figure 5: Cumulative Return of A2C and the DOW index

This visualization does not provide concrete metrics for the performance of the algorithms, but can be used to gain insight into the nature of the predictions. For example, during the maximum drawdown period A2C encountered during March, the algorithm bridged the gap avoiding the lowest prices. The graph also provides an indication of how much better or worse the algorithm is performing compared to the baseline. With further testing, the algorithms will be compared to other established funds such as the S&P 500, and NASDAQ 100.

The above tests were all shown on one version on the A2C algorithm. These tests will be applied to all algorithms and parameters tested to be compared with one another. To ensure a better understanding of the algorithms other metrics, such as rolling volatility and monthly distribution of returns, will also be computed. Additionally, the algorithms will be tested on in-sample data to check for over-fitting.

6 CONCLUSION

Work on this project is already underway, with a section of Objective 1 already being achieved. Going forward the project will focus on completing all the objectives by the dates depicted in the gantt chart (See appendix A). Each objective has been allocated an adequate amount of time for its completion.

In the coming days, progress on Objective 1 will continue with a focused on hyperparameter optimization, model training and model Evaluation. Objective 1 has been allocated the largest amount of time due to its complexity. Development of Objective 2 shall begin on the 20th January with the creation and testing of the ensemble strategy. The objective has been allocated 37 days for completion. Objective three follows a similarly approach with 26 days to apply the ensembles strategy to a live trading environment. Report writing will be conducted throughout the development of the project.

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7 APPENDIX A

Thesis

Objective 1

Hyper Parameter Optimization
Model Training
Model and Parameter Evaluation
Fine Tuning

Objective 2

Creation of Ensemble Strategy
Strategy Testing
Refinement

Objective 3

Trading Platform Research
Paper Trading Implementation

Report

Report Writing and Notes
Report Refinement and Finalization
Review Period
Submission

